

Development Setup

Source: docs/development-setup.md

Development Setup

Status: internal technical documentation Last updated: 2026-05-23 Extended reference: DEVELOPER_GUIDE.md, DEPLOYMENT_GUIDE.md

The local reference environment is Windows + PowerShell. The backend can also be run from other shells if paths are adjusted.

Prerequisites

Tool	Purpose
Python 3.11+	FastAPI, Celery tasks and tests
Node.js 18+	Vite/React frontend
Docker + Docker Compose	Redis, Celery worker/beat, Flower
Git	Version control

The backend dependencies are listed in backend/requirements.txt. Important packages include fastapi, uvicorn, celery, redis, openai, supabase, python-docx, pdfplumber, openpyxl, reportlab, pytest and python-jose[cryptography].

Environment Variables

Do not commit .env.

Required or commonly used variables:

```
[powershell]
OPENAI_API_KEY=sk-...
SUPABASE_URL=https://<project>.supabase.co
SUPABASE_SERVICE_ROLE_KEY=<service-role-key>
DATABASE_URL=postgresql://...
REDIS_URL=redis://localhost:6379/0
JWT_SECRET=<random-secret-min-32-chars>
JWT_ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=60
SIGNUP_ENABLED=false
TZ=Europe/Warsaw
```

Optional frontend variables:

```
[powershell]
VITE_API_URL=http://localhost:8000
VITE_REPORT_MODEL_LABEL=AI POWERED
```

The current embedding model is text-embedding-3-small. Report and chat generation use gpt-4o-mini.

Backend

From the repository root:

```
[powershell]
.\.venv\Scripts\Activate.ps1
$env:PYTHONPATH="."
python -m uvicorn backend.main:app --reload --host 0.0.0.0 --port 8000
```

The API runs at http://localhost:8000; OpenAPI docs are available at http://localhost:8000/docs.

Redis, Celery and Flower

```
[powershell]
docker compose up -d redis celery-worker celery-beat
docker compose --profile monitoring up -d flower
```

Useful logs:

```
[powershell]
docker compose logs -f celery-worker
```

Frontend

```
[powershell]
cd frontend
npm.cmd install
npm.cmd run dev
npm.cmd run build
npm.cmd run lint
```

The Vite dev server runs at `http://localhost:5173`. There is no Vite proxy in `frontend/vite.config.js`; frontend API calls use `VITE_API_URL` and fall back to `http://localhost:8000`.

Use `npm.cmd`, not `npm`, on Windows PowerShell if `npm.ps1` is blocked by ExecutionPolicy.

Fast Backend Tests

From the repository root:

```
[powershell]
.\.venv\Scripts\Activate.ps1
$env:PYTHONPATH="."
python -m pytest backend\test_common_endpoints.py backend\test_negative_integration.py
python -m pytest backend\test_pdf_generator.py backend\test_report_tasks.py
python -m pytest backend\test_rag_quality_sample_docs.py
```

The RAG quality sample test is offline and deterministic. It parses `sample_uploads/*.docx`, uses the real chunker and monkeypatches embedding/RPC calls so it does not require OpenAI, Supabase or Celery.

E2E Report Test

Run only when API, Redis/Celery, OpenAI, Supabase and an admin/admin account are available:

```
[powershell]
python backend\test_e2e.py Environmental
```

RAG Diagnostics

```
[powershell]
python diagnose_rag.py "pytanie" "<user_id>" "Environmental"
```

The script calls the same retrieval path as the app and prints matched chunks, source headers and a user-doc vs knowledge-base split.

Common Issues

Celery task stays PENDING

Check that Redis is running and `REDIS_URL` matches backend and worker config.

OpenAI embedding calls fail

Check `OPENAI_API_KEY` and access to `text-embedding-3-small`.

PDF has broken Polish characters

Install or configure TTF fonts. The PDF generator tries Arial, DejaVu Sans, Liberation Sans and ReportLab Vera before falling back to Helvetica.

Frontend cannot reach backend

Set `VITE_API_URL=http://localhost:8000` or rely on the same fallback. The Vite config does not proxy API calls.